

HERNANDO, PANTINE B. CHALLENGE - 17 - CRYPTOGRAPHY

BSIT-3

This challenge is a classic introduction to substitution ciphers, specifically the **Vigenère cipher**. While the description mentions a "one-time pad," the use of a repeating key and a lookup table (the Vigenère square) makes this a standard polyalphabetic substitution puzzle.

Easy1 

 

Medium Cryptography picoCTF 2019

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Hints ?

Description

1 2

The one time pad can be cryptographically secure, but not when you know the key. Can you solve this?

We've given you the encrypted flag, key, and a table to help UFJKXQZQUNB with the key of SOLVECRYPTO. Can you use this [table](#) to solve it?.

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Challenge Breakdown

- **Ciphertext:** UFJKXQZQUNB
- **Key:** SOLVECRYPTO
- **Tools Provided:** A "table" (Vigenère Square/Tabula Recta).

VIEW

Plaintext

CRYPTOISFUN

ENCODE DECODE

Vigenère cipher

VARIANT

Standard Vigenère cipher

KEY

SOLVECRYPTO

KEY MODE

Repeat

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY

Maintain case

FOREIGN CHARS

Include Ignore

Decoded 13 chars

VIEW

Ciphertext

UFJKXQZQUNB

Methodology

The Vigenère cipher works by shifting each letter of the plaintext by the value of the corresponding letter in the key. To decrypt it manually using the table:

1. Find the row corresponding to the first letter of the **Key** (**S**).
2. Look across that row until you find the first letter of the **Ciphertext** (**U**).
3. Move up to the top of that column to find the original **Plaintext** letter.

Decryption Table

Ciphertext	Key	Process (Shift Calculation)	Plaintext
U (20)	S (18)	$(20 - 18) \bmod 26 = 2$	C
F (5)	O (14)	$(5 - 14) \bmod 26 = 17$	R
J (9)	L (11)	$(9 - 11) \bmod 26 = 24$	Y

K (10)	V (21)	$(10 - 21) \bmod{26} = 15$	P
X (23)	E (4)	$(23 - 4) \bmod{26} = 19$	T
Q (16)	C (2)	$(16 - 2) \bmod{26} = 14$	O
Z (25)	R (17)	$(25 - 17) \bmod{26} = 8$	I
Q (16)	Y (24)	$(16 - 24) \bmod{26} = 18$	S
U (20)	P (15)	$(20 - 15) \bmod{26} = 5$	F
N (13)	T (19)	$(13 - 19) \bmod{26} = 20$	U
B (1)	O (14)	$(1 - 14) \bmod{26} = 13$	N

Decrypted String: CRYPTOISFUN

Final Flag

picoCTF{CRYPTOISFUN}

Reflection

This challenge serves as a great reminder that **terminology matters**. Calling this a "one-time pad" is technically correct only if the key is truly random, never reused, and kept secret. In the context of a CTF where the key is provided in the description and follows a readable English pattern (SOLVECRYPTO), it essentially functions as a Vigenère cipher.

From a security standpoint, it highlights the "Key Distribution Problem." A cipher can be mathematically unbreakable (like a true One-Time Pad), but if the key is compromised

or easily guessed, the encryption becomes trivial to bypass. For future challenges, whenever a "table" is mentioned alongside a key of equal length to the ciphertext, the Tabula Recta should be the first tool you reach for.